

Alan Amadeus Davila

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Education

Texas A&M University — College Station, TX

B.S. in Electrical & Computer Engineering | Expected Graduation: December 2026

Professional Development: LinkedIn Learning — *Become a PLC Developer* Learning Path, 2026

Projects

AC-to-DC Regulated Power Supply

Independent Project, 2026

- Built a low-voltage AC-to-DC regulated power supply to demonstrate transformer isolation, bridge rectification, filtering, and voltage regulation principles.
- Tested 12VAC input, rectified DC bus voltage, and regulated 5VDC output using DMM and oscilloscope measurements under varying load conditions.
- Documented schematic, waveform captures, ripple behavior, and troubleshooting process to validate power conversion and output stability.

Low-Voltage DC-to-AC Converter

Independent Project, 2026

- Designed a low-voltage DC-to-AC converter using Arduino-controlled H-bridge switching to generate an AC-style waveform from a 12V DC source.
- Tested output frequency, duty cycle, and waveform behavior using oscilloscope and DMM measurements under different load conditions.
- Documented switching performance, load response, and design limitations related to square-wave inversion and power conversion efficiency.

Arduino UNO R4 WiFi Closed-Loop Drawer Controller [\[GitHub\]](#)

Independent Project, 2026

C++, Arduino, Quadrature Encoders, IBT-2 Motor Drivers, Limit Switches

- Designed an upgraded **Arduino UNO R4 WiFi control architecture** using dual quadrature encoders and four end-of-travel limit switches for position feedback and boundary detection.
- Developed **C++ closed-loop positioning logic** to command each drawer to 0%, 50%, or 100% based on encoder counts instead of fixed motor run times.
- Implemented automatic homing, endpoint protection, and safety timeouts to establish reference positions and prevent mechanical overtravel.

Technical Skills:

Power & Electrical: AC/DC conversion, DC/AC conversion, rectification, voltage regulation, filtering, H-bridge switching, PWM, load testing, analog/digital circuits

Test & Measurement: Oscilloscope, DMM, waveform analysis, frequency response, data acquisition

Software & Tools: LTspice, Multisim, MATLAB/Simulink, AutoCAD, Arduino IDE, Vivado, Git, Linux

Programming: C, C++, Python, Verilog, SystemVerilog, VHDL

Recent Work Experience

Texas A&M University - College Station

August 2022 – Present

Junior Facilities Coordinator

College Station, TX

- Inspected completed campus maintenance work orders using AggieWorks DB, verifying contractor/internal work completion and documenting deficiencies requiring follow-up service.
- Coordinated with university facilities staff, SCC personnel, third-party contractors, and BAS/facilities personnel to track repair status and support timely completion of building maintenance tasks.